Chapter 3. Techniques of Proof

Given a true mathematical statement, we are most interested in how we can verify that it is true. A *proof* provides such an explanation or verification. In this chapter we shall study various techniques of mathematical proof. Before the discussion, let us first describe briefly the meanings of a few common terms used in mathematics books:

- **Definition**: an explanation of the meaning of a term.
- **Theorem** (or **Proposition**): an important and interesting true mathematical statement.
- **Lemma**: a true mathematical statement used in verifying other true statements; usually considered to be less important than a theorem, and can be regarded as a "helping result".
- **Corollary**: a true mathematical statement that is a simple deduction from a theorem.
- **Proof**: an explanation of why a statement is true.
- **Conjecture**: a statement that is believed to be true, but for which we have no proof.

3.1. Direct Proof

Most theorems can be stated as implications in the form $\forall x \in S,\ (P(x) \Rightarrow Q(x))$, where $P(x)$ and $Q(x)$ are open statements concerning elements x in some set S. This means $\forall x \in S,$ "$P(x) \to Q(x)$" is true. In general practice, however, we usually omit the phrase "is true", and simply express the theorem as "For any $x \in S$, if $P(x)$, then $Q(x)$", or sometimes "Let $x \in S$. If $P(x)$, then $Q(x)$". For example, the statement

"The square of any odd integer is odd"

can be expressed as "For any $x \in \mathbb{Z}$, (x is odd $\Longrightarrow x^2$ is odd)", or "For any $x \in \mathbb{Z}$, if x is odd, then x^2 is odd", or "Let $x \in \mathbb{Z}$. If x is odd, then x^2 is odd".

The first method we introduce to prove statements of the above form is known as the method of *direct proof*. Suppose that $P(x_0)$ is false for some $x_0 \in S$. Then the conditional $P(x_0) \to Q(x_0)$ is automatically true for this element x_0. It follows that there is no need to consider those $x \in S$ for which $P(x)$ is false. Thus to give a direct proof of $\forall x \in S,\ (P(x) \Rightarrow Q(x))$, we will first assume that $P(x)$ is true, and then proceed to show that $Q(x)$ is true.

Example 3.1 Prove that for any $n \in \mathbb{Z}$, if n is even, then $5n + 3$ is odd.

Proof. We would like to show that $\forall n \in \mathbb{Z},\ (P(n) \Rightarrow Q(n))$, where $P(n)$ means "n is even" and $Q(n)$ means "$5n + 3$ is odd". Assume that $P(n)$ is true, that is, assume that n is an even integer. First we need to recall the fact that every odd integer can be written as $2k + 1$ for some $k \in \mathbb{Z}$ (and vice versa), whereas every even integer can be written as $2k$ for some $k \in \mathbb{Z}$ (and vice versa).

Since we have assumed that n is even, so $n = 2k$ for some $k \in \mathbb{Z}$. Now

$$5n + 3 = 5(2k) + 3 = 10k + 3 = 2(5k + 1) + 1 = 2k' + 1 \quad \text{where } k' = 5k + 1 \in \mathbb{Z}.$$

Since $5n + 3$ can be written as $2k' + 1$ for some $k' \in \mathbb{Z}$, we conclude that $5n + 3$ is odd. □

Example 3.2 Prove that for any $n \in \mathbb{Z}$, if n is odd, then $3n^2 + n - 1$ is odd.

Proof. Assume that n is odd. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$. Now

$$\begin{aligned} 3n^2 + n - 1 &= 3(2k + 1)^2 + (2k + 1) - 1 = 3(4k^2 + 4k + 1) + 2k = 12k^2 + 14k + 3 \\ &= 2(6k^2 + 7k + 1) + 1 = 2k' + 1 \end{aligned}$$

where $k' = 6k^2 + 7k + 1 \in \mathbb{Z}$. Hence we conclude that $3n^2 + n - 1$ is odd. □

Example 3.3 Prove that for any $x \in \mathbb{R}$, if $0 < x < 2$, then $(x - 1)^2 < 1$.

Proof. Write $P(x) : 0 < x < 2$ and $Q(x) : (x - 1)^2 < 1$. Assume that $0 < x < 2$. Then we have

$$x^2 - 2x = \underbrace{x}_{\text{positive}} \underbrace{(x - 2)}_{\text{negative}} < 0.$$

Therefore, $(x - 1)^2 = (x^2 - 2x) + 1 < 1$. □

Example 3.4 Prove that for any $x \in \mathbb{R}$, if $(x - 1)^2 < 1$, then $0 < x < 2$.

Proof. Assume that $(x-1)^2 < 1$. Then we have $x^2 - 2x + 1 - 1 < 0$, that is, $x(x-2) < 0$. Therefore, either

$$x < 0 \quad \text{and} \quad (x - 2) > 0$$

or

$$x > 0 \quad \text{and} \quad (x - 2) < 0.$$

It is impossible to find a real number x satisfying the first one. The second one gives $0 < x < 2$. □

Combining Example 3.3 and 3.4 we actually obtain an "if and only if" result:

$$\text{For any } x \in \mathbb{R}, \quad \text{we have } (x - 1)^2 < 1 \ \text{ if and only if } \ 0 < x < 2.$$

Recall that $P(x) \Leftrightarrow Q(x)$ means $P(x) \Rightarrow Q(x)$ and $Q(x) \Rightarrow P(x)$. Thus to give a direct proof of $P(x) \Leftrightarrow Q(x)$, we need to break the proof into (i) $P(x) \Rightarrow Q(x)$ and (ii) $Q(x) \Rightarrow P(x)$. (i) is known as the "*only if*" part (the $\Rightarrow$ direction); (ii) is the *converse* of (i) and is known as the "*if*" part (the $\Leftarrow$ direction).

Remark In set language, the equivalence between $P(x)$ and $Q(x)$ suggests that

$$\{x \in \mathbb{R} : 0 < x < 2\} = \{x \in \mathbb{R} : (x - 1)^2 < 1\}.$$

Proofs involving Congruence of Integers

We know that an integer x can be expressed as $2k$ or $2k+1$ for some integer k, according to whether the remainder is 0 or 1 when x is divided by 2. Hence if two integers x and y have the same remainder when divided by 2, then their difference $x-y$ is a multiple of 2. Similarly, an integer x can be expressed as $3k$, $3k+1$, or $3k+2$ for some integer k, according to whether the remainder is 0, 1 or 2 when x is divided by 3. Hence if two integers x and y have the same remainder when divided by 3, then their difference $x-y$ is a multiple of 3. This leads us to the following definition:

Definition 3.5 For any $a, b \in \mathbb{Z}$ and for any $n \in \mathbb{N}$ with $n \geq 2$, we say that $a \equiv b \pmod{n}$, read as *a is congruent to b modulo n*, if $a-b$ is a multiple of n. This means $a-b=nk$ for some $k \in \mathbb{Z}$.

For example, $19 \equiv 7 \pmod{4}$ since $19-7$ is a multiple of 4; $-15 \equiv 5 \pmod{5}$ since $-15-5$ is a multiple of 5. On the other hand, 16 is not congruent to 8 modulo 6, written $16 \not\equiv 8 \pmod{6}$, since $16-8$ is not a multiple of 6.

We now consider some results of congruence of integers and their proofs:

Example 3.6 Prove that for any $a, b, c, d \in \mathbb{Z}$ and for any $n \in \mathbb{N}$ with $n \geq 2$, if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a+c \equiv b+d \pmod{n}$.

Proof. Assume that $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Then there exist $k_1, k_2 \in \mathbb{Z}$ such that

$$a-b=nk_1 \quad \text{and} \quad c-d=nk_2.$$

Hence

$$(a+c)-(b+d)=(a-b)+(c-d)=nk_1+nk_2=n(k_1+k_2),$$

and therefore $a+c \equiv b+d \pmod{n}$ as k_1+k_2 is also an integer. □

For example, we have $2 \equiv 7 \pmod{5}$ and $3 \equiv 8 \pmod{5}$, then the above result shows that $5 \equiv 15 \pmod{5}$.

Example 3.7 Prove that for any $x \in \mathbb{Z}$, if $x \equiv 1 \pmod{3}$, then $x^2 \equiv 1 \pmod{3}$.

Proof. Assume that $x \equiv 1 \pmod{3}$. Then there exists $k \in \mathbb{Z}$ such that $x-1=3k$. Hence $x^2-1=(x-1)(x+1)=3k(x+1)$, and therefore $x^2 \equiv 1 \pmod{3}$ as $k(x+1)$ is also an integer. □

Exercise

(i) Prove that for any $a, b, c \in \mathbb{Z}$ and for any $n \in \mathbb{N}$ with $n \geq 2$, if $a \equiv b \pmod{n}$, then $ca \equiv cb \pmod{n}$.

(ii) Prove that for any $a, b \in \mathbb{Z}$ and for any $n \in \mathbb{N}$ with $n \geq 2$, if $a \equiv b \pmod{n}$, then $a^2 \equiv b^2 \pmod{n}$.

(iii) Prove that for any $a, b, c, d \in \mathbb{Z}$ and for any $n \in \mathbb{N}$ with $n \geq 2$, if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $ac \equiv bd \pmod{n}$.

(iv) Prove that for any $x \in \mathbb{Z}$, if $x \equiv 2 \pmod{3}$, then $x^2 \equiv 1 \pmod{3}$.

Remark For any integer x and any positive integer n, according to the *Division Algorithm*[1], x has remainder either 0 or 1 or 2 or $\cdots$ or $n-1$ when it is divided by n. This means for any integer x, we have $x \equiv 0 \pmod{n}$ or $1 \pmod{n}$ or $2 \pmod{n}$ or $\cdots$ or $n-1 \pmod{n}$.

Proofs involving Sets

We now discuss proofs involving sets. Recall that a set X is a subset of another set Y means every element in X is an element in Y. Thus to prove $X \subseteq Y$, we need to prove $\forall x$, $(x \in X \Rightarrow x \in Y)$, that is, for any x, if $x \in X$, then $x \in Y$.

Example 3.8 Prove that for any sets A, B and C, if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Proof. Assume that $A \subseteq B$ and $B \subseteq C$. Our goal is to show that $A \subseteq C$, that is, we want to show that for any x, if $x \in A$, then $x \in C$. The proof proceeds as follows.

For any x, if $x \in A$, then $x \in B$ since $A \subseteq B$ by assumption. Since $B \subseteq C$, it follows that $x \in C$. Thus we have shown that for any x, if $x \in A$, then $x \in C$, that is, $A \subseteq C$. □

Remark The following way of presenting the above proof is also acceptable:

Proof. For any x, we have

$$\begin{aligned} x \in A &\implies x \in B \quad (\text{since } A \subseteq B) \\ &\implies x \in C \quad (\text{since } B \subseteq C). \end{aligned}$$

□

To show the equality of two sets X and Y, we need to prove that $X \subseteq Y$ and $Y \subseteq X$. Furthermore, many results concerning sets follow from corresponding results about logical statements that were discussed in *Chapter 2*.

Example 3.9 Prove that for any sets A, B and C, $A \cup (B \cup C) = (A \cup B) \cup C$.
(cf. *Theorem 1.19 (c)*: associative law for union of sets)

Proof. We have to prove that (i) $A \cup (B \cup C) \subseteq (A \cup B) \cup C$ and (ii) $(A \cup B) \cup C \subseteq A \cup (B \cup C)$.

To show (i), for any x, if $x \in A \cup (B \cup C)$, we have $x \in A$ or ($x \in B$ or $x \in C$). This statement is logically equivalent to ($x \in A$ or $x \in B$) or $x \in C$ by the associative law of disjunction (*Example 2.7 (v)*). Thus $x \in (A \cup B) \cup C$, and hence $A \cup (B \cup C) \subseteq (A \cup B) \cup C$. The proof of (ii) is similar and is therefore omitted. □

[1] The Division Algorithm states that for any integer b and for any positive integer a, there exist unique integers q and r such that $b = qa + r$ and $0 \leq r < a$.

Remark In general, if we find that the proof of the other inclusion can be achieved by just reversing the order of the arguments given in the previous part, then we can combine the proof of two inclusions together.

Example 3.10 Prove that for any sets A, B and C, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
(cf. *Theorem 1.23 (a)*: distributive law)

Proof. This time we prove the two inclusions simultaneously. According to the distributive law in logic (*Example 2.7 (vii)*), we have $P \wedge (Q \vee R)$ is logically equivalent to $(P \wedge Q) \vee (P \wedge R)$. Thus for any x,

$$\begin{aligned} x \in A \cap (B \cup C) &\iff (x \in A) \text{ and } ((x \in B) \text{ or } (x \in C)) \\ &\iff ((x \in A) \text{ and } (x \in B)) \text{ or } ((x \in A) \text{ and } (x \in C)) \\ &\iff x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

□

Example 3.11 Let A and B be subsets of a universal set U. Prove that $(A \cup B)^c = A^c \cap B^c$.
(cf. *Theorem 1.28*: De Morgan's law)

Proof. According to the De Morgan's law in logic (*Example 2.7 (ix)*), we have $\sim (P \vee Q)$ is logically equivalent to $(\sim P) \wedge (\sim Q)$. Thus for any x,

$$\begin{aligned} x \in (A \cup B)^c &\iff x \in U \text{ and } (x \notin A \cup B) \\ &\iff x \in U \text{ and } ((x \notin A) \text{ and } (x \notin B)) \\ &\iff x \in A^c \text{ and } x \in B^c \\ &\iff x \in A^c \cap B^c. \end{aligned}$$

□

Example 3.12 Let A and B be subsets of a universal set U. Prove that $A \setminus B = A \cap B^c$. Hence show that $(A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A)$.

Proof. Note that for any x,

$$x \in A \setminus B \iff (x \in A \text{ and } x \notin B) \iff (x \in A \text{ and } x \in B^c) \iff x \in A \cap B^c.$$

Thus we have proved $A \setminus B = A \cap B^c$. Next,

$$\begin{aligned} (A \cup B) \setminus (A \cap B) &= (A \cup B) \cap (A \cap B)^c \quad \text{(by the previous result)} \\ &= (A \cup B) \cap (A^c \cup B^c) \quad \text{(De Morgan's Law)} \\ &= [(A \cup B) \cap A^c] \cup [(A \cup B) \cap B^c] \quad \text{(Distributive Law)} \\ &= [(A \cap A^c) \cup (B \cap A^c)] \cup [(A \cap B^c) \cup (B \cap B^c)] \quad \text{(Distributive Law)} \\ &= [\emptyset \cup (B \cap A^c)] \cup [(A \cap B^c) \cup \emptyset] \\ &= (B \cap A^c) \cup (A \cap B^c) \\ &= (B \setminus A) \cup (A \setminus B) \quad \text{(by the previous result)} \end{aligned}$$

□

We finish this section by looking at some examples of proofs involving Cartesian products of sets.

Example 3.13 Prove that for any sets A, B, C and D, if $A \subseteq C$ and $B \subseteq D$, then $A \times B \subseteq C \times D$.

Proof. For any (x, y),

$$\begin{aligned}(x,y) \in A \times B &\implies x \in A \text{ and } y \in B \\ &\implies x \in C \text{ and } y \in D \quad (\text{since } A \subseteq C \text{ and } B \subseteq D) \\ &\implies (x,y) \in C \times D.\end{aligned}$$

□

Example 3.14 Prove that for any sets A, B and C, $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

Proof. Note that for any (x, y),

$$\begin{aligned}(x,y) \in A \times (B \cup C) &\iff x \in A \text{ and } y \in B \cup C \\ &\iff x \in A \text{ and } (y \in B \text{ or } y \in C) \\ &\iff (x \in A \text{ and } y \in B) \text{ or } (x \in A \text{ and } y \in C) \quad (\text{Distributive Law}) \\ &\iff (x,y) \in A \times B \text{ or } (x,y) \in A \times C \\ &\iff (x,y) \in (A \times B) \cup (A \times C)\end{aligned}$$

Thus we have proved $A \times (B \cup C) = (A \times B) \cup (A \times C)$. □

Example 3.15 Prove that for any sets A, B and C, $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.

Proof. Note that for any (x, y),

$$\begin{aligned}(x,y) \in A \times (B \setminus C) &\implies x \in A \text{ and } y \in B \setminus C \\ &\implies x \in A \text{ and } y \in B \text{ and } y \notin C \\ &\implies (x,y) \in A \times B \text{ and } (x,y) \notin A \times C \quad (\text{since } y \notin C) \\ &\implies (x,y) \in (A \times B) \setminus (A \times C).\end{aligned}$$

This proves $A \times (B \setminus C) \subseteq (A \times B) \setminus (A \times C)$. Moreover, for any (x, y),

$$\begin{aligned}(x,y) \in (A \times B) \setminus (A \times C) &\implies (x \in A \text{ and } y \in B) \text{ and } (x \notin A \text{ or } y \notin C) \\ &\implies (x \in A \text{ and } y \in B \text{ and } x \notin A) \text{ or } (x \in A \text{ and } y \in B \text{ and } y \notin C) \\ &\implies \mathbf{F} \text{ or } (x \in A \text{ and } y \in B \text{ and } y \notin C) \\ &\implies (x \in A \text{ and } y \in B \text{ and } y \notin C) \\ &\implies (x,y) \in A \times (B \setminus C).\end{aligned}$$

This completes the proof. □

Exercise 3.16 Prove that for any sets A, B and C,

(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(ii) $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$

(iii) $(A \cap B) \setminus C = (A \setminus C) \cap (B \setminus C)$

(iv) $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$

(v) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

(vi) $A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$

(vii) if $A \subseteq B$, then $C \setminus B \subseteq C \setminus A$.

3.2. Proof by Contrapositive

Given a conditional $P \to Q$, its contrapositive is the statement $\sim Q \to \sim P$ which is logically equivalent to $P \to Q$. This means if we can show that $\sim Q \Rightarrow \sim P$, then it follows that $P \Rightarrow Q$. A *proof by contrapositive* of $\forall x \in S$, $P(x) \Rightarrow Q(x)$ is a direct proof of $\forall x \in S, \sim Q(x) \Rightarrow \sim P(x)$, that is, we assume that $\sim Q(x)$ is true and then proceed to show that $\sim P(x)$ is true. In other words, we assume that $Q(x)$ is false and then proceed to show that $P(x)$ is false. There are some situations where a proof by contrapositive is more preferable. Let us take a look at some examples:

Example 3.17 Prove that for any $n \in \mathbb{Z}$, if $5n+3$ is odd, then n is even.

Proof. Let $P(n)$ be the statement "$P(n)$: $5n+3$ is odd", and $Q(n)$ be the statement "$Q(n)$: n is even". We wish to show that $\forall n \in \mathbb{Z}, P(n) \Rightarrow Q(n)$. Here we are going to apply the method of proof by contrapositive, that is, we want to show $\forall n \in \mathbb{Z}, \sim Q(n) \Rightarrow \sim P(n)$.

Assume that $\sim Q(n)$ is true, that is, assume that n is an odd integer. Then $n = 2k+1$ for some $k \in \mathbb{Z}$. Now

$$5n+3 = 5(2k+1)+3 = 10k+8 = 2(5k+4) = 2k' \quad \text{where } k' = 5k+4 \in \mathbb{Z}.$$

Since $5n+3$ can be written as $2k'$ for some $k' \in \mathbb{Z}$, we conclude that $5n+3$ is even. □

Remarks

(i) Example 3.17 is the converse of Example 3.1. Combining the two results gives "For any $n \in \mathbb{Z}$, n is even if and only if $5n+3$ is odd".

(ii) If we had attempted to prove Example 3.17 with a direct proof, then we would have to assume that $5n+3$ is odd. On the other hand, applying the method of proof by contrapositive allows us to work with n initially instead of working with the more complicated expression $5n+3$.

Example 3.18 Prove that for any $n \in \mathbb{Z}$, n is even if and only if n^2 is even.

Proof. The "only if" part (the $\Rightarrow$ direction): Assume that n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$, and we have

$$n^2 = 4k^2 = 2(2k^2).$$

Since $2k^2$ is an integer, it follows that n^2 is even.

The "if" part (the $\Leftarrow$ direction): We need to show that if n^2 is even, then n is even. Here we are going to apply the method of proof by contrapositive, as this allows us to work with n initially rather than n^2. This means we have to prove that if n is odd, then n^2 is odd.

If n is odd, then $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Hence n^2 is odd as $2k^2 + 2k$ is an integer. □

Remark The statement in Example 3.18 is logically equivalent to the statement

"For any $n \in \mathbb{Z}$, n is odd if and only if n^2 is odd."

Exercise

(i) Prove that for any $n \in \mathbb{Z}$, n is odd if and only if $5n - 11$ is even.

(ii) Prove that for any $n \in \mathbb{Z}$, n^3 is even if and only if n is even.

3.3. Proof by Cases

Often we have to break down a problem into several cases and tackle each case individually. It may sometimes be useful to further divide a case into subcases. Some typical cases are given by: (i) an integer n is even or odd; (ii) a real number x is less than 0, equal to 0 or greater than 0; (iii) for two nonzero real numbers x and y, either $xy > 0$ or $xy < 0$. For each of the cases in (iii), it may be further divided into two subcases:

$$xy > 0 \quad \begin{cases} x > 0 \text{ and } y > 0 \\ x < 0 \text{ and } y < 0 \end{cases} \qquad xy < 0 \quad \begin{cases} x > 0 \text{ and } y < 0 \\ x < 0 \text{ and } y > 0 \end{cases}$$

Example 3.19 Prove that for any $n \in \mathbb{Z}$, $n^2 + 3n + 6$ is even.

Proof. We can divide the proof into two cases: n is even and n is odd.

<u>Case 1.</u> If n is even, then $n = 2k$ for some $k \in \mathbb{Z}$. Hence

$$n^2 + 3n + 6 = (2k)^2 + 3(2k) + 6 = 4k^2 + 6k + 6 = 2(2k^2 + 3k + 3).$$

Therefore $n^2 + 3n + 6$ is even as $2k^2 + 3k + 3$ is an integer.

<u>Case 2.</u> If n is odd, then $n = 2k + 1$ for some $k \in \mathbb{Z}$. Hence

$$n^2 + 3n + 6 = (2k+1)^2 + 3(2k+1) + 6 = 4k^2 + 4k + 1 + 6k + 3 + 6 = 2(2k^2 + 5k + 5).$$

Therefore $n^2 + 3n + 6$ is even as $2k^2 + 5k + 5$ is an integer.

It follows that $n^2 + 3n + 6$ is even for any integer n. □

Example 3.20 Prove that for any $m, n \in \mathbb{Z}$, m and n are of the same parity if and only if $m+n$ is even.

Proof. Two integers are said to be *of the same parity* if they are both even or both odd; otherwise they are *of different parity*.

First we give the proof for the "only if" part. Assume that m and n are of the same parity. There are two cases to be considered:

Case 1. Both m and n are even. Then $m = 2k$ and $n = 2k'$ for some integers k, k'. Hence $m + n = 2k + 2k' = 2(k + k')$. Therefore $m + n$ is even as $k + k'$ is an integer.

Case 2. Both m and n are odd. Then $m = 2k + 1$ and $n = 2k' + 1$ for some integers k, k'. Hence $m + n = (2k + 1) + (2k' + 1) = 2(k + k' + 1)$. Therefore $m + n$ is even as $k + k' + 1$ is an integer.

Next we give the proof for the "if" part. Here we are going to apply the method of proof by contrapositive. Assume the negation of "m and n are of the same parity", that is, assume that m and n are of different parity. Again, there are two cases to be considered:

Case 1. If m is even and n is odd. Then $m = 2k$ and $n = 2k' + 1$ for some integers k, k'. Hence $m + n = 2k + (2k' + 1) = 2(k + k') + 1$. Therefore $m + n$ is odd as $k + k'$ is an integer.

Case 2. If m is odd and n is even. The proof of this case is similar to that of the previous one and is therefore omitted. □

Remark Sometimes we may use the phrase *Without Loss of Generality (WLOG)* to indicate that the proofs of the two situations are similar, and hence the proof of only one of these is needed. For example, in the "if" part of the above proof, we may write "Assume that m and n are of different parity. WLOG, assume that m is even and n is odd". Then we can proceed with the proof for this case without any further comment on the other case.

Example 3.21 (Triangle Inequality) For any $x, y \in \mathbb{R}$, $|x + y| \leq |x| + |y|$.

Proof. As the absolute value function is a case-defined function and it depends on the sign of the variable, it is therefore natural to divide the proof into cases depending on whether x and y are both positive, both negative, or differ in sign.

Case 1. Suppose that $x \geq 0$ and $y \geq 0$. Then $x + y \geq 0$, and hence

$$|x + y| = x + y = |x| + |y|.$$

Case 2. Suppose that $x < 0$ and $y < 0$. Then $x + y < 0$, and hence

$$|x + y| = -(x + y) = (-x) + (-y) = |x| + |y|.$$

Case 3. Suppose that one of x and y is nonnegative and the other negative. WLOG we may assume that $x \geq 0$ and $y < 0$. There are two subcases to consider: $x + y \geq 0$ and $x + y < 0$.

Subcase 1. Suppose that $x + y \geq 0$. Then

$$\begin{aligned} |x+y| &= x+y \\ &< x+(-y) \quad \text{as } y < 0 < -y \\ &= |x|+|y|. \end{aligned}$$

Subcase 2. Suppose that $x + y < 0$. Then

$$\begin{aligned} |x+y| &= -(x+y) \\ &= (-x)+(-y) \\ &\leq x+(-y) \quad \text{as } -x \leq 0 \leq x \\ &= |x|+|y|. \end{aligned}$$

Hence the inequality is true for any $x, y \in \mathbb{R}$. □

3.4. Proof by Contradiction

In this section we discuss another very important proof technique called *proof by contradiction*.

Let S be a statement. Suppose we are going to prove that S is true. If we can show that the conditional $\sim S \to$ **F** is true (that is, $\sim S \Rightarrow$ **F**), where **F** denotes a contradiction, then the truth table for conditional tells us that $\sim S$ must be false. Hence S must be true, which is what we want to show.

In other words, we first assume that the statement we want to prove to be true is false and proceed to show that this gives a contradiction. Thus our assumption must be wrong, which means it is impossible for the statement to be false, as it leads to something ridiculous, and hence the statement is true. This technique is known as *proof by contradiction*. The name comes from the fact that assuming that the statement is false is later contradicted by some other fact. It is also known as *reductio ad absurdum* which means reduction to absurdity.

Example 3.22 Prove that no odd number can be written as a sum of two odd numbers.

Proof. Let S be the statement "No odd number can be written as a sum of two odd numbers." Assume to the contrary that S is false, that is, assume that there is an odd number m which can be written as a sum of two odd numbers x and y. Write $x = 2k + 1$ and $y = 2k' + 1$ for some integers k and k'. Then we have

$$m = x + y = (2k + 1) + (2k' + 1) = 2(k + k' + 1).$$

Hence m is even, as $k + k' + 1$ is an integer. Now we arrive at a contradiction as m is given as an odd number. Thus our assumption that "there is an odd number which is a sum of two odd numbers" must be wrong, that is, no odd number is a sum of two odd numbers. □

Remark: Compare this result with Example 3.20.

Example 3.23 Prove that there is no smallest positive real number.

Proof. Assume to the contrary that there is a smallest positive real number x. Consider the number $x/2$. Since x is positive, so is $x/2$. But we know that $0 < x/2 < x$, and so $x/2$ is a positive real number which is less than x. This contradicts with our assumption that x is the smallest positive real number. Hence the assumption is false, and we have proved that there is no smallest positive real number. □

Remark: The results in Example 3.22 and 3.23 are so-called "negative-sounding" results. Proofs of these negative-sounding results are often given by the method of proof by contradiction.

Example 3.24 Prove that there are infinitely many prime numbers.

Proof. Assume to the contrary that there are only finitely many prime numbers, say there are n prime numbers. We list them in ascending order as follows: $p_1 < p_2 < \ldots < p_n$. Consider the number

$$q = p_1 p_2 \ldots p_n + 1.$$

Clearly, we have $q \geq 2$. By the *Fundamental Theorem of Arithmetic*[2], there is a prime number dividing q (it is possible that q itself is a prime). Since $p_1, p_2, \ldots, p_n$ are all the primes by our assumption, so one of them, say p_i, must divide q. As p_i clearly divides $p_1 p_2 \cdots p_n$, it follows that p_i divides $q - p_1 p_2 \cdots p_n$, that is, p_i divides 1. This is certainly impossible, and hence we arrive at a contradiction. Thus our assumption is false and we have shown that there are infinitely many prime numbers. □

Remark: The first known proof for the above result is usually attributed to Euclid. Many more proofs are now known, see for instance, Chapter 1 – Six proofs of the infinity of primes, *Proofs from THE BOOK* by M. Aigner and G.M. Ziegler, Springer, 4th edition.

Example 3.25 Prove that the number $\sqrt{2}$ is irrational, that is, $\sqrt{2}$ cannot be written in the form p/q for any $p, q \in \mathbb{Z}$ with $q \neq 0$.

Proof. Assume to the contrary that $\sqrt{2}$ is rational. Then $\sqrt{2} = p/q$ for some $p, q \in \mathbb{Z}$ with $q \neq 0$. By dividing p and q by any common factors greater than or equal to 2, if necessary, we may further assume

[2] The Fundamental Theorem of Arithmetic says that every integer greater than or equal to two is either a prime or can be expressed as a product of primes.

that p and q have no common factor greater than or equal to 2, that is, p/q has been expressed in (or reduced to) lowest terms. Then we have $2 = p^2/q^2$ and so $p^2 = 2q^2$. This implies that p^2 is even as q^2 is an integer. Example 3.18 then suggests that p must be even, and hence we have $p = 2k$ for some integer k. Substituting this into the equation $p^2 = 2q^2$ gives $4k^2 = 2q^2$. Therefore we have $q^2 = 2k^2$. This means q^2 is even as k^2 is an integer. By Example 3.18 again, we have q is even. Since both p and q are even, they have 2 as a common factor, and so we arrive at a contradiction as p/q has been reduced to lowest terms. Hence our assumption that $\sqrt{2}$ is rational is false and so $\sqrt{2}$ must be irrational. □

To show that statements of the form "$\forall x \in S,\ P(x) \to Q(x)$" is true by using a proof by contradiction, we assume that "$\exists x \in S, \sim (P(x) \to Q(x))$" is true and proceed to deduce a contradiction. But $\sim (P(x) \to Q(x))$ is logically equivalent to $P(x) \wedge (\sim Q(x))$. Thus to prove $\forall x \in S, P(x) \Rightarrow Q(x)$ by contradiction, we assume that there exists $x \in S$ such that $P(x)$ is true and $Q(x)$ is false, and then deduce a contradiction.

Example 3.1 (re-visit) Prove that for any $n \in \mathbb{Z}$, if n is even, then $5n + 3$ is odd.

Proof. Here we use a proof by contradiction. Assume that there exists $n \in \mathbb{Z}$ such that n is even and $5n + 3$ is even. Since n is even, we write $n = 2k$ for some $k \in \mathbb{Z}$. Then

$$5n + 3 = 5(2k) + 3 = 10k + 3 = 2(5k + 1) + 1.$$

Hence $5n+3$ is odd as $5k+1$ is an integer. But this is a contradiction as $5n+3$ is both odd and even. □

Example 3.26 Prove that for any $x \in \mathbb{R} \setminus \{0\}$, if $x + \frac{1}{x} < 2$, then $x < 0$.

Proof. Assume that there exists $x \in \mathbb{R} \setminus \{0\}$ such that $x + \frac{1}{x} < 2$ and $x \geq 0$. Since x is nonzero, we must have $x > 0$. Multiplying both sides of $x + \frac{1}{x} < 2$ by x gives $x^2 + 1 < 2x$. It follows that $x^2 - 2x + 1 < 0$, and so we have $(x - 1)^2 < 0$. This is a contradiction. □

We finish this section by giving a remark on how proof by contradiction is different from proof by contrapositive. When we are trying to prove $P \Rightarrow Q$ by the method of contradiction, we assume P and $\sim Q$ are both true and show that a contradiction results; in the method of contrapositive, we assume $\sim Q$ is true and prove that $\sim P$ is true. The method of contrapositive has the advantage that the goal is clear; in the method of contradiction, the goal is to prove a contradiction, but it is not always obvious what the contradiction is going to be, and so the goal is less clear.

3.5. Existence Proof and Disproof by Counterexample

Existence Proof

Recall that a statement with the existential quantifier is of the form $\exists x \in S,\ P(x)$ where S is some set, and $P(x)$ is an open statement. This statement is true if $P(x)$ is true for at least one $x \in S$; otherwise this is false. Hence to prove such a statement, we need only to find an element $x \in S$ that makes $P(x)$ true. This is known as an *existence proof*.

Example 3.27 Prove that there exist integers a, b, c such that $a^2 + b^2 = c^2$.

Proof. The proof is simple in this case, as we can take

$$a = 3, \quad b = 4, \quad c = 5.$$

Then

$$a^2 + b^2 = 3^2 + 4^2 = 5^2 = c^2. \qquad \square$$

Example 3.28 Prove that there exist integers a, b such that $a^3 + b^3 = (a + b)^3$.

Proof. Again the proof is simple, as we can take

$$a = 1, \quad b = 0.$$

Then

$$a^3 + b^3 = 1^3 + 0^3 = 1^3 = (a + b)^3. \qquad \square$$

Example 3.29 Prove that there exist irrational numbers a and b such that a^b is rational.

Proof. Previously we have shown that $\sqrt{2}$ is irrational. Consider the number $\sqrt{2}^{\sqrt{2}}$. This number is either rational or irrational. We consider the following two cases:

<u>Case 1.</u> $\sqrt{2}^{\sqrt{2}}$ is rational. We then take $a = b = \sqrt{2}$ and we are done.

<u>Case 2.</u> $\sqrt{2}^{\sqrt{2}}$ is irrational. We then take $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$. Both a and b are irrational. Now,

$$a^b = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = (\sqrt{2})^{(\sqrt{2}\cdot\sqrt{2})} = (\sqrt{2})^2 = 2,$$

which is rational.

Therefore no matter $\sqrt{2}^{\sqrt{2}}$ is rational or not, we can always find two irrational numbers a and b such that a^b is rational. $\square$

Remark This proof may seem unsatisfactory as we still do not know two specific irrational numbers a and b such that a^b is rational. This is known as a *non-constructive proof* as it does not construct an example a and b. However, whether $\sqrt{2}^{\sqrt{2}}$ is rational or not does not affect the conclusion (it turns out that $\sqrt{2}^{\sqrt{2}}$ is irrational!).

Disproof by Counterexample

Suppose we are going to show that the statement "$\forall x \in S,\ P(x)$" is false, that is, we wish to show that $\exists x \in S$ such that $P(x)$ is false. An element $a \in S$ for which $P(a)$ is false is known as a *counterexample* of this statement.

To *disprove* the statement "$\forall x \in S,\ P(x)$", we mean to prove that it is false. Thus a single counterexample is all we need to disprove the statement.

Example 3.30 Disprove that $x^2 - 1 \geq 0$ for any $x \in \mathbb{R}$.

Solution. This means we need to find an $x \in \mathbb{R}$ such that $x^2 - 1 < 0$. Clearly, $x = 1/2$ serves the purpose. □

Example 3.31 Disprove that the product of any two irrational numbers is irrational.

Solution. This means we need to find $x, y \in \mathbb{I}$ such that $xy \in \mathbb{Q}$. Take $x = \sqrt{2}$ and $y = \sqrt{2}$. Then both x and y are irrational, but $xy = 2$ is clearly rational. □

If the statement is stated in the form "$\forall x \in S, P(x) \to Q(x)$, then one may disprove this by showing that there exists some element $a \in S$ such that $P(a) \to Q(a)$ is false, that is, $P(a)$ is true and $Q(a)$ is false.

Example 3.32 Disprove that for any $x \in \mathbb{Z}$, if $x^2 \equiv 1 \pmod 3$, then $x \equiv 1 \pmod 3$.

Solution. We need to find an integer x such that $x^2 \equiv 1 \pmod 3$ and $x \not\equiv 1 \pmod 3$. Take $x = 5$ and we have $25 \equiv 1 \pmod 3$ but $5 \not\equiv 1 \pmod 3$. □

Now suppose we are asked to disprove the statement "$\exists x \in S,\ P(x)$". We then need to show that no such an x exists so that $P(x)$ is true, that is, we have to verify that "$\forall x \in S, \sim P(x)$" is true. This requires us to give a proof of $\forall x \in S, \sim P(x)$, instead of constructing counterexamples.

Example 3.33 Disprove that there exists $n \in \mathbb{Z}$ such that $n^2 + 3n + 6$ is odd.

Solution. This means we need to show that $n^2 + 3n + 6$ is even for any $n \in \mathbb{Z}$. But we have already given the proof in Example 3.19. □

We finish this section by providing some problems in which the truth or falseness of each of the statements is not specified. We are required to determine whether each statement is true or not, then prove it if it is true and disprove it if it is false.

Exercise Prove or disprove each of the following statements:

(a) There exists a real number x satisfying the equation $x^8 + 3x^2 + 1 = 0$.

(b) Let A, B and C be sets. If $A \times C = B \times C$, then $A = B$.

(c) If x and y are positive real numbers, then $|x + y| = |x| + |y|$.

(d) There exists a real number x such that $x^3 < x < x^2$.

(e) $\forall x \in \mathbb{I}$ and $x > 0$, $\exists y \in \mathbb{I}$, $0 < y < x$.

(f) There exist odd numbers x and y such that $x^2 + 3y^2 \equiv 0 \pmod 6$.

3.6. Proof by Mathematical Induction

The last method we introduce in this chapter is known as *proof by mathematical induction*. This method is applied to establish the truth of a statement of the form "For any positive integer n greater than or equal to m, $P(n)$ is true."

Suppose that we have a sequence of statements $P(m), P(m+1), P(m+2), \ldots$ which we would like to show are true. We can certainly show that each of these is true individually. *The Principle of Mathematical Induction* provides an alternative method:

Theorem 3.34 (The Principle of Mathematical Induction) *For any $m \in \mathbb{Z}$, if $P(m)$, $P(m+1)$, $P(m+2)$, $\ldots$ are statements satisfying*

(i) $P(m)$ is true; and

(ii) for any integer $k \geq m$, $P(k) \Longrightarrow P(k+1)$,

then the statements $P(m),\ P(m+1),\ P(m+2), \ldots$ are true.

Thus to show that the statement $P(n)$ is true for any $n \geq m$, we need only to prove two things, namely (i) $P(m)$ is true and (ii) for any integer $k \geq m$, $P(k) \Rightarrow P(k+1)$. This is a two-step process. The verification of (i) is called the *basis step* or *base case*. To prove the implication in (ii) is known as the *inductive step*. For this purpose, we often use a direct proof and assume that $P(k)$ is true for an arbitrary integer k with $k \geq m$. This assumption is called the *inductive (or induction) hypothesis*.

We shall omit the proof of the principle of mathematical induction which follows from the *Well-Ordering Principle*.[3] Let us illustrate the method of proof by mathematical induction with some examples:

Example 3.35 Prove that for any $n \in \mathbb{N}$,

$$P(n): \ 1+2+3+\cdots+n = \frac{n(n+1)}{2}$$

is true.

Proof. We are required to show that $P(1)$, $P(2)$, $P(3)$, ... are all true. Let us prove this by mathematical induction. Here we take $m = 1$ in Theorem 3.34.

Basis Step: The statement $P(1)$ is certainly true, since we have $1 = \frac{1(1+1)}{2}$. This establishes the basis step.

Inductive Hypothesis: Let k be an arbitrary integer such that $k \geq 1$. Assume that $P(k)$ is true, that is, assume that

$$1+2+3+\cdots+k = \frac{k(k+1)}{2}$$

is true for this particular k.

Inductive Step: We wish to prove that $P(k+1)$ is true. Observe that

$$\begin{aligned} 1+2+3+\cdots+k+(k+1) &= \frac{k(k+1)}{2}+(k+1) \quad \text{(by the inductive hypothesis)} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This verifies that $P(k+1)$ is true. Hence by the principle of mathematical induction, $P(n)$ is true for every positive integer n. □

Remarks:

(i) In the inductive hypothesis of our proof in Example 3.35, we assume that $P(k)$ is true for an arbitrary positive integer k. We certainly do not assume that $P(k)$ is true for *all* positive integers k, for this is precisely what we wish to prove.

[3] The Well-Ordering Principle states that the set $\mathbb{N}$ of natural numbers is well-ordered. See for instance Section 6.1 of *Mathematical Proofs: A Transition to Advanced Mathematics (2nd edition)* for more details.

(ii) There are other ways of showing the result in Example 3.35. However, in general, mathematical induction provides a technique that can be applied to a wide range of problems.

Example 3.36 Prove that for any $n \in \mathbb{N}$ with $n \geq 5$,

$$P(n): \ 2^n > n^2$$

is true.

Proof. We are required to show that $P(5)$, $P(6)$, $P(7)$, $\ldots$ are all true. This time we take $m = 5$ in Theorem 3.34.

Basis Step: The statement $P(5)$ is certainly true, since we have $2^5 = 32 > 25 = 5^2$. This establishes the basis step.

Inductive Hypothesis: Let k be an arbitrary integer such that $k \geq 5$. Assume that $P(k)$ is true, that is, assume that

$$2^k > k^2$$

is true for this particular k.

Inductive Step: We wish to prove that $P(k+1)$ is true, that is, $2^{k+1} > (k+1)^2$ is true. Observe that

$$2^{k+1} = 2 \times 2^k > 2k^2 \qquad \text{(by the inductive hypothesis)}$$

and

$$2k^2 - (k+1)^2 = k^2 - 2k - 1 = k(k-2) - 1 > 0 \quad \text{as } k \geq 5.$$

This verifies that $P(k+1)$ is true. Hence by the principle of mathematical induction, $P(n)$ is true for every integer $n \geq 5$. □

Example 3.37 Prove that for any set A and for any integer $n \geq 0$, if A is finite with $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.

Proof. Denote by A_n a finite set with cardinality n. Then we are required to prove that $P(n)$: $|\mathcal{P}(A_n)| = 2^n$ is true for every integer $n \geq 0$.

Basis Step: The statement $P(0)$ is certainly true, since A_0 is indeed the empty set $\emptyset$, and $\mathcal{P}(\emptyset)$ is the set $\{\emptyset\}$ which has $1 = 2^0$ element. This establishes the basis step.

Inductive Hypothesis: Let k be an arbitrary integer such that $k \geq 0$. Assume that $P(k)$ is true, that is, $|\mathcal{P}(A_k)| = 2^k$ is true for this particular k. This means that there are 2^k subsets of A_k.

Inductive Step: We wish to prove that $P(k+1)$ is true. First fix an element $a \in A_{k+1}$. Let $B = A_{k+1} \setminus \{a\}$. Then the set B contains k elements and hence, by the inductive hypothesis, there are 2^k subsets of B.

Let S be a subset of B. Then S is also a subset of A_{k+1}. Note that S does not include the element a as $a \notin B$. Then $S \cup \{a\}$ will also be a subset of A_{k+1}. It follows that every subset S of B gives rise to two subsets of A_{k+1}, namely, S itself and $S \cup \{a\}$. Consequently, there are 2^k subsets of A_{k+1} *without* a and 2^k subsets of A_{k+1} *with* a. Altogether we have $2^k + 2^k$ subsets of A_{k+1}. Therefore, $|\mathcal{P}(A_{k+1})| = 2^{k+1}$. This verifies that $P(k+1)$ is true. Hence by the principle of mathematical induction, $P(n)$ is true for every integer $n \geq 0$. □

The Strong Form of Induction

In the last part of this section we are going to introduce a stronger form of mathematical induction. In some cases, assuming just $P(k)$ is true is not enough to deduce that $P(k+1)$ is true, we need to use the fact that the property is true for all cases from the basis step to the given k. This strengthening of the induction hypothesis produces the following *Strong Form of Induction*:

Theorem 3.38 (The Strong Form of Induction) *For any $m \in \mathbb{Z}$, if $P(m)$, $P(m+1)$, $P(m+2)$, $\ldots$ are statements satisfying*

(i) $P(m)$ is true; and

(ii) for any integer $k \geq m$, $P(m) \wedge \cdots \wedge P(k) \Longrightarrow P(k+1)$,

then the statements $P(m)$, $P(m+1)$, $P(m+2), \ldots$ are true.

Thus to show that the statement $P(n)$ is true for any $n \geq m$ by the strong form of induction, we need to prove (i) the basis step: $P(m)$ is true, and (ii) the inductive step: for any integer $k \geq m$, if $P(m)$, $P(m+1)$, $\ldots$, $P(k-1)$, $P(k)$ are true, then $P(k+1)$ is true.

Example 3.39 Prove that if a sequence $x_1, x_2, x_3, \ldots$ of numbers satisfying $x_1 = 1$, $x_2 = 3$ and

$$x_n = 2x_{n-1} - x_{n-2} \quad \text{for any integer } n \geq 3$$

then $x_n = 2n - 1$ for any $n \in \mathbb{N}$.

Proof. Let $P(n)$ be the statement $P(n) : x_n = 2n - 1$. We shall apply the strong form of induction to prove that $P(1)$, $P(2)$, $P(3)$, $\ldots$ are all true. We take $m = 1$ in Theorem 3.38.

Basis Step: The statement $P(1)$ is true, since we have $x_1 = 1 = 2 \cdot 1 - 1$. This establishes the basis step.

Inductive Hypothesis: Let k be an arbitrary integer such that $k \geq 1$. Assume that $P(1), \ldots, P(k)$ are true, that is, assume that $x_i = 2i - 1$ for all integers i with $1 \leq i \leq k$.

Inductive Step: We wish to show that $P(k+1)$ is true, that is, $x_{k+1} = 2(k+1)-1 = 2k+1$, where k is an arbitrary integer with $k \geq 1$. When $k = 1$, we have $x_2 = 3 = 2 \cdot 1 + 1$ and so $P(k+1) : x_{k+1} = 2k+1$ is true for $k = 1$. Now when $k \geq 2$, that is, $k + 1 \geq 3$, we can apply the given *recurrence relation*:

$$x_{k+1} = 2x_k - x_{k-1}.$$

Then $x_{k+1} = 2x_k - x_{k-1} = 2(2k-1) - (2k-3) = 2k+1$ by the inductive hypothesis. It follows that $P(k+1)$ is true. Hence by the strong form of induction, $P(n)$ is true for any $n \in \mathbb{N}$. □

Example 3.40 Prove that if a sequence $x_1, x_2, x_3, \ldots$ of numbers satisfying $x_1 = 1$, $x_2 = 2$, $x_3 = 3$ and

$$x_n = x_{n-1} + x_{n-2} + x_{n-3} \quad \text{for any integer } n \geq 4$$

then $x_n < 2^n$ for any $n \in \mathbb{N}$.

Proof. Let $P(n)$ be the statement $P(n) : x_n < 2^n$. We shall apply the strong form of induction to prove that $P(1)$, $P(2)$, $P(3)$, $\ldots$ are all true. We take $m = 1$ in Theorem 3.38.

Basis Step: The statement $P(1)$ is certainly true, since we have $x_1 = 1 < 2^1$. This establishes the basis step.

Inductive Hypothesis: Let k be an arbitrary integer such that $k \geq 1$. Assume that $P(1), \ldots, P(k)$ are true, that is, assume that $x_i < 2^i$ for all integers i with $1 \leq i \leq k$.

Inductive Step: We wish to show that $P(k+1)$ is true, that is, $x_{k+1} < 2^{k+1}$, where k is an arbitrary integer with $k \geq 1$. When $k = 1, 2$, we have $x_2 = 2 < 2^2$ and $x_3 = 3 < 2^3$, so $P(k+1) : x_{k+1} < 2^{k+1}$ is true when $k = 1, 2$. Now when $k \geq 3$, that is, $k + 1 \geq 4$, we can apply the given recurrence relation:

$$x_{k+1} = x_k + x_{k-1} + x_{k-2}.$$

Then

$$x_{k+1} = x_k + x_{k-1} + x_{k-2} < 2^k + 2^{k-1} + 2^{k-2} = 2^{k-2}(2^2 + 2 + 1) < 2^{k-2} \cdot 8 = 2^{k+1}$$

by the inductive hypothesis. It follows that $P(k+1)$ is true. Hence by the strong form of induction, $P(n)$ is true for any $n \in \mathbb{N}$. □